

The logo of K. J. Somaiya College of Engineering is a shield-shaped emblem. It features a central tower with a flag, flanked by two smaller structures. Below the tower are various symbols including a gear, a wrench, a screwdriver, and a lightbulb. The text "K. J. SOMAIYA COLLEGE OF ENGINEERING" is inscribed across the middle of the shield.

Experiment No.: 02

Title: Git and GitHub

Aim: Study Git, GitHub and upload an application on GitHub.

Resources needed: GitHub Account, Git installation

Theory:

Introduction to Git

Git is a version control system for tracking changes in computer files and coordinating work on those files among multiple people. It is primarily used for source code management in software development, but it can be used to keep track of changes in any set of files.

Git Commands

Git Init

The git init command creates a new Git repository. It can be used to convert an existing, unversioned project to a Git repository or initialize a new, empty repository.

```
C:\Users\admin\git-demo-project>git init
Initialized empty Git repository in C:/Users/admin/git-demo-project/.git/
```

Git Add

When we create, modify or delete a file, these changes will happen in our local and won't be included in the next commit (unless we change the configurations). We need to use the git add command to include the changes of a file(s) into our next commit. The git add command doesn't change the repository and the changes are not saved until we use git commit. The git add command adds new or changed files in your working directory.

```
C:\Users\admin\git-demo-project>git add stack.py

C:\Users\admin\git-demo-project>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   stack.py
```

Git Commit

A commit, or "revision", is an individual change to a file (or set of files). It's like when you save a file, except with Git, every time you save it creates a unique ID (a.k.a. the "SHA" or "hash") that allows you to keep record of what changes were made when and by who. In version control systems, a commit is an operation which sends the latest changes to the source code to the repository, making these changes part of the head revision of the repository.

```
C:\Users\admin\git-demo-project>git commit -m "Initial Commit"
[master (root-commit) 0b2f679] Initial Commit
1 file changed, 41 insertions(+)
create mode 100644 stack.py
```

Git-Push and Pull

Push operation: The git push command is used to upload local repository content to a remote repository. Pushing is how you transfer commits from your local repository to a remote repository. The command for push is `git push remote -all`.

Pull operation: The git pull command is used to fetch and download content from a remote repository and immediately update the local repository to match that content. The command is: `git pull <remote>`.

Git Clone

Git clone is a Git command line utility which is used to target an existing repository and create a clone, or copy of the target repository. In this page we'll discuss extended configuration options and common use cases of git clone. Clones a repository into a newly created directory, creates remote-tracking branches for each branch in the cloned repository (visible using `git branch --remotes`), and creates and checks out an initial branch that is forked from the cloned repository's currently active branch.

Branches: Git branches are effectively a pointer to a snapshot of your changes. When you want to add a new feature or fix a bug, no matter how big or how small, you spawn a new branch to encapsulate your changes. This makes it harder for unstable code to get merged into the main code base, and it gives you the chance to clean up your future's history before merging it into the main branch.

Procedure:

1. Download Git tool from official Git website: <https://git-scm.com/downloads>
 2. Create GitHub Account <https://github.com/join>
 3. Link Git with GitHub with credentials
 4. Create a folder on GitHub
 5. Clone the folder
 6. Execute Git Commands: version, global, init, add, commit, checkout, push, pull, clone, branch, merge etc.
 7. Write any sample web application and expose it on localhost.
 8. Push application on GitHub
-

Results: (Document with screenshots)

1. Application code
2. Pushed file on GitHub dashboard

```
C:\Windows\system32\cmd.e  x  +  v

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>git config --global user.name "chandana-galgali"

C:\Users\Dell>git config --global user.email "chandana.galgali@gmail.com"

C:\Users\Dell>ssh-keygen -t rsa -b 4096 -C "chandana.galgali@gmail.com"
Generating public/private rsa key pair.
Enter file in which to save the key (C:\Users\Dell\.ssh\id_rsa):
Created directory 'C:\Users\Dell\.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Dell\.ssh\id_rsa
Your public key has been saved in C:\Users\Dell\.ssh\id_rsa.pub
The key fingerprint is:
SHA256:u/+hsl0oYDRi/fIVZJB051nG00PXzW5mBK98XT7TdPE chandana.galgali@gmail.com
The key's randomart image is:
+---[RSA 4096]-----+
|  . . o               |
|  + B o . .          |
|  = + . . = o        |
|  o * . o . = E      |
|  . o o .So + . +o    |
|  + . . .o * .oo     |
|  . + .o = . o       |
|  o * . .            |
|  .+. =o.            |
+---[SHA256]-----+

Activate Windows
Go to Settings to activate Windows.
```

SSH and GPG keys

github.com/settings/keys

Settings

Chandana Galgali (chandana-galgali)

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Organizations

Enterprises

Moderation

Code, planning, and automation

Repositories

Codespaces

Models

SSH keys

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

chandana.galgali@gmail.com

SHA256:u/+hsl0oYDRi/fIVZJB051nG00PXzW5mBK98XT7TdPE

Added on Aug 1, 2025

Never used — Read/write

Delete

Check out our guide to [connecting to GitHub using SSH keys](#) or [troubleshoot common SSH problems](#).

GPG keys

There are no GPG keys associated with your account.

Learn how to [generate a GPG key and add it to your account](#).

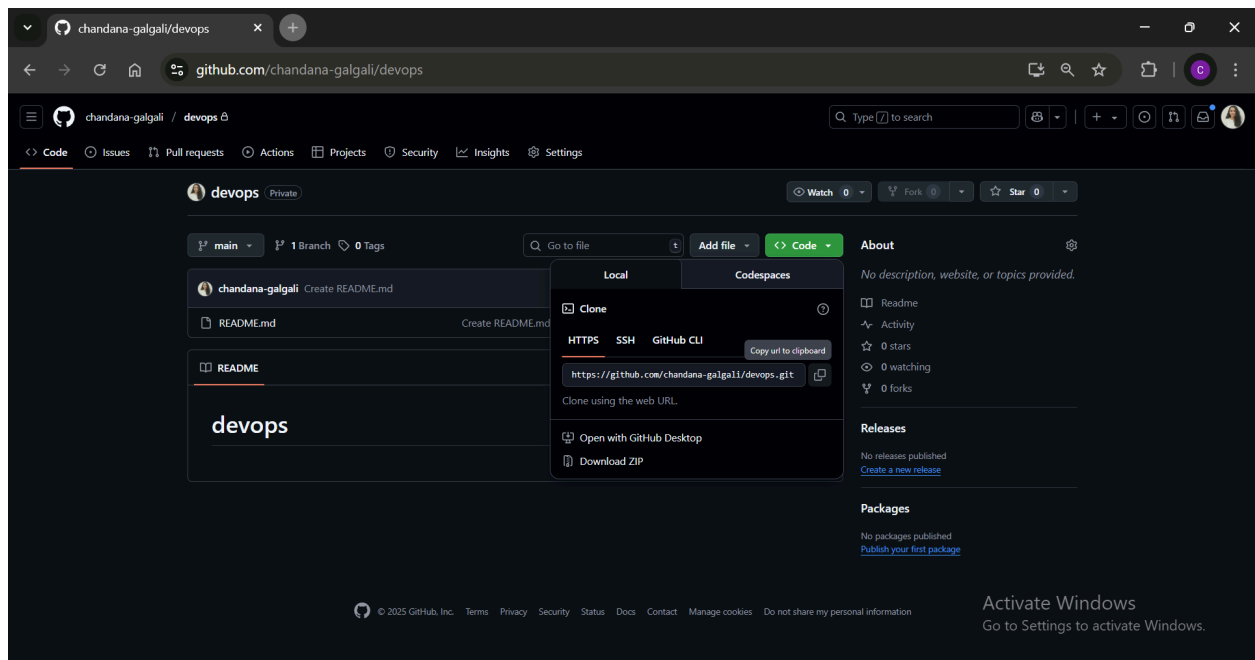
Vigilant mode

☐ Flag unsigned commits as unverified

This will include any commit attributed to your account but not signed with your GPG or S/MIME key. Note that this will include your existing unsigned commits. [Learn about vigilant mode](#).

Activate Windows

Go to Settings to activate Windows.



```
C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2>git clone https://github.com/chandana-galgali/devops.git
Cloning into 'devops'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2>cd devops

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git init
Reinitialized existing Git repository in C:/Users/Dell/Downloads/VII SEM/DEVOPS/LAB/EXP-2/devops/.git/
```

```
C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>echo "This is the first line of text in file.txt" > file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>dir
Volume in drive C has no label.
Volume Serial Number is E4F5-FAA1

Directory of C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops

01-08-2025  22:25    <DIR>          .
01-08-2025  22:17    <DIR>          ..
01-08-2025  22:25                47 file.txt
01-08-2025  22:17                10 README.md
                2 File(s)              57 bytes
                2 Dir(s)  37,843,300,352 bytes free

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git add file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git commit -m "Created file.txt and added initial text"
[main 1a72528] Created file.txt and added initial text
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 340 bytes | 340.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/chandana-galgali/devops.git
   05a098b..1a72528  main -> main
```

Activate Windows
Go to Settings to activate Windows.

```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch new-feature-branch

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main
  new-feature-branch

```

```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git checkout new-feature-branch
Already on 'new-feature-branch'

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>dir
Volume in drive C has no label.
Volume Serial Number is E4F5-FAA1

Directory of C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops

01-08-2025  22:25    <DIR>          .
01-08-2025  22:17    <DIR>          ..
01-08-2025  22:25                47 file.txt
01-08-2025  22:17                10 README.md
                2 File(s)              57 bytes
                2 Dir(s) 37,844,602,880 bytes free

```

```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>echo "This is the first line of text in feature_file.txt" > feature_file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git add feature_file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch new-feature-branch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   feature_file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git commit -m "Created feature_file.txt and added initial text"
[new-feature-branch d921813] Created feature_file.txt and added initial text
 1 file changed, 1 insertion(+)
 create mode 100644 feature_file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch new-feature-branch
nothing to commit, working tree clean

```

```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git merge new-feature-branch
Updating 1a72528..d921813
Fast-forward
 feature_file.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature_file.txt

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git push origin new-feature-branch
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 388 bytes | 388.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'new-feature-branch' on GitHub by visiting:
remote:   https://github.com/chandana-galgali/devops/pull/new/new-feature-branch
remote:
To https://github.com/chandana-galgali/devops.git
 * [new branch]      new-feature-branch -> new-feature-branch

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main
  new-feature-branch

```

```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main
  new-feature-branch

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/chandana-galgali/devops.git
   1a72528..d921813  main -> main

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

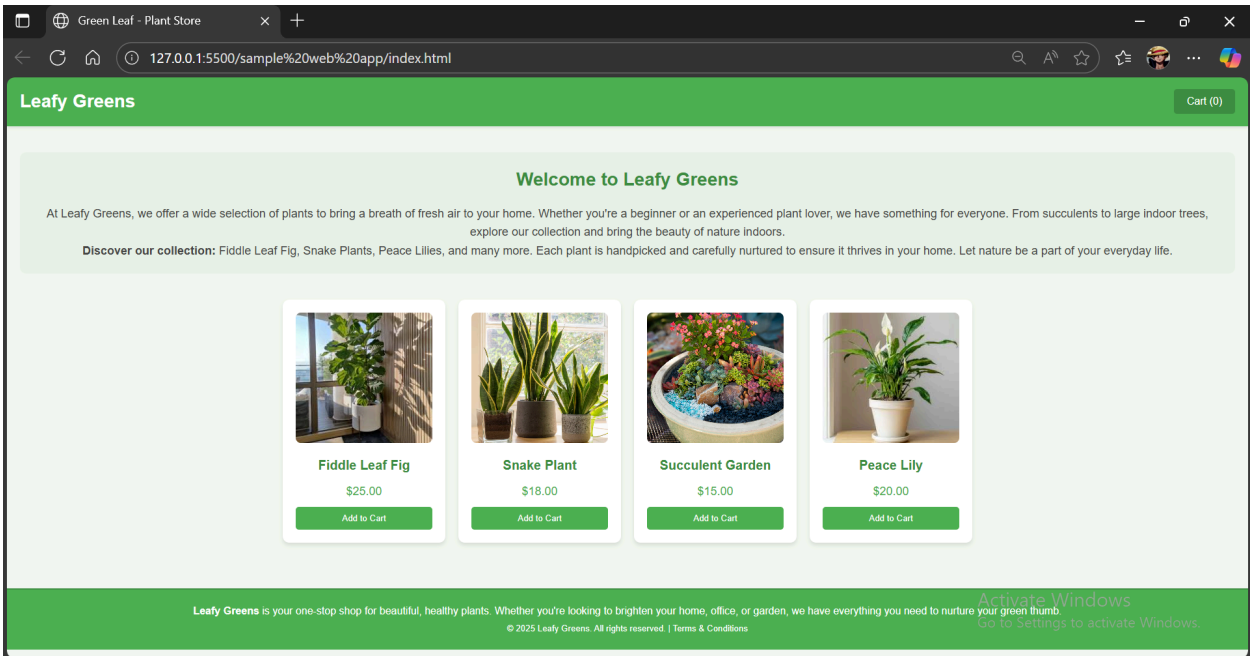
C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main
  new-feature-branch

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch -d new-feature-branch
Deleted branch new-feature-branch (was d921813).

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git push origin --delete new-feature-branch
To https://github.com/chandana-galgali/devops.git
   - [deleted]      new-feature-branch

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git branch
* main

```



```

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git add .

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   sample web app/index.html
    new file:   sample web app/plant1.jpg
    new file:   sample web app/plant2.jpg
    new file:   sample web app/plant3.jpg
    new file:   sample web app/plant4.jpg
    new file:   sample web app/script.js
    new file:   sample web app/style.css

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git commit -m "Initial commit of sample web app"
[main 8e3ba5e] Initial commit of sample web app
 7 files changed, 357 insertions(+)
 create mode 100644 sample web app/index.html
 create mode 100644 sample web app/plant1.jpg
 create mode 100644 sample web app/plant2.jpg
 create mode 100644 sample web app/plant3.jpg
 create mode 100644 sample web app/plant4.jpg
 create mode 100644 sample web app/script.js
 create mode 100644 sample web app/style.css

```

```
C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 544.81 KiB | 2.42 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/chandana-galgali/devops.git
   d921813..8e3ba5e  main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EXP-2\devops>git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

The screenshot shows the GitHub repository page for 'devops' by user 'chandana-galgali'. The page displays the 'sample web app' directory with a list of files: index.html, plant1.jpg, plant2.jpg, plant3.jpg, plant4.jpg, script.js, style.css, README.md, feature_file.txt, and file.txt. The commit history shows the initial commit of the sample web app, dated 5 minutes ago.

The screenshot shows the GitHub repository page for 'devops' by user 'chandana-galgali'. The page displays the 'devops' directory with a list of files: README.md, feature_file.txt, and file.txt. The commit history shows the initial commit of the sample web app, dated 8 minutes ago. The page also displays the 'About' section with a description, activity, stars, watching, and forks.

Questions:

1. What is the difference between Git and GitHub?

Git is a distributed version control system used for tracking changes in source code and coordinating work among developers. It allows for local version control on a developer's machine. GitHub is a cloud-based hosting service for Git repositories. It provides a web interface for managing Git repositories, collaborative features like pull requests, issue tracking, and continuous integration services.

2. What is Master and Branch in Github?

The Master branch in Git (now commonly referred to as the Main branch) is the default branch created when a new repository is initialized. It often represents the production-ready version of the code. A Branch is a pointer to a snapshot of your changes. Developers use branches to work on new features or bug fixes without affecting the main codebase. Once the work on a branch is completed, it can be merged back into the main branch.

Outcomes: CO3 – Comprehend the effective code building and testing process

Conclusion: (Conclusion to be based on the Results and outcomes achieved)

In this experiment, we learned how to use Git and GitHub for version control and collaboration in software development. Git allows tracking code changes, while GitHub provides a platform for hosting and sharing repositories. We initialized a Git repository, created branches, made changes, and pushed them to GitHub. We also cloned a repository from GitHub, demonstrating local and remote repository synchronization. This hands-on experience helped us understand version control workflows and the importance of branches for collaboration.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

References:

Books/ Journals/ Websites:

1. https://www.w3schools.com/git/git_install.asp?remote=github
 2. <https://opensource.com/article/18/1/step-step-guide-git>
 3. Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale, Jennifer Davis, Ryn Daniels, O'Reilly Media June 2016.
 4. Practical DevOps: Implement DevOps in your organization by effectively building, deploying, testing, and monitoring code,Joakim Verona, Packt Publishing, 2nd Edition,2016
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